Lab 7: Evaluating and Controlling Processes in RHEL 9

Section 1: Get Information About Running Processes

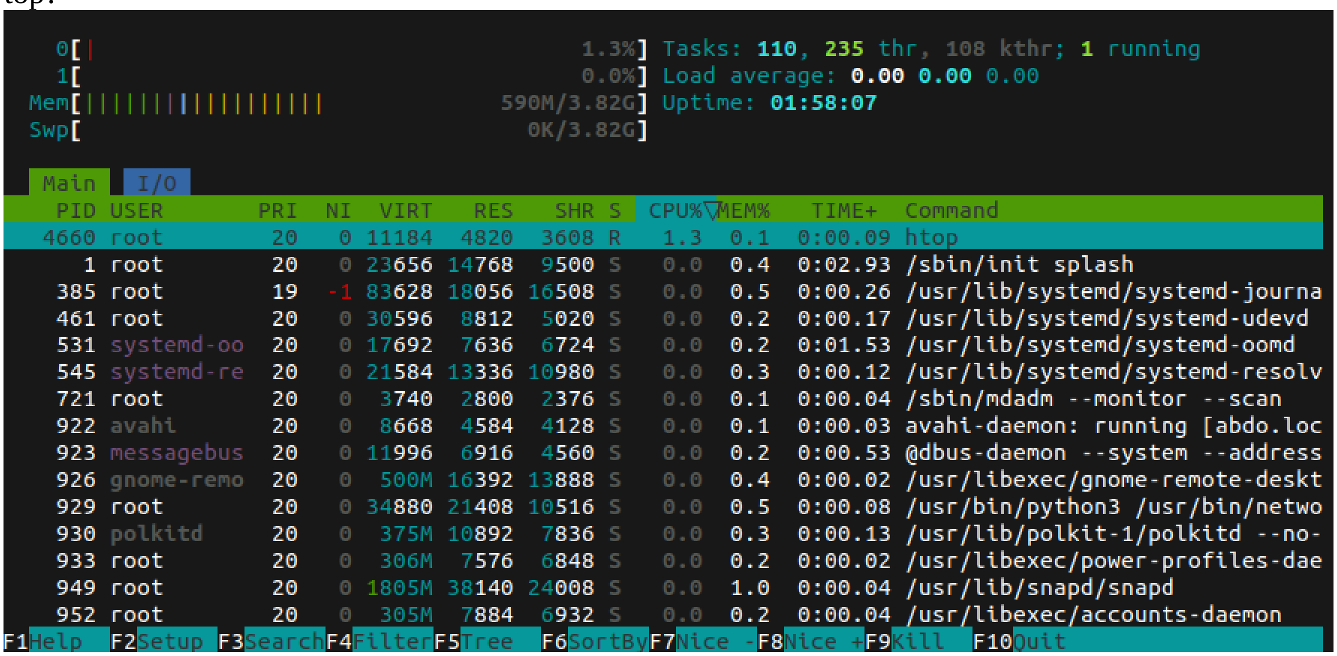
Question: List all running processes on the system using the ps command. Which information does this command display for each process?

Answer : **ps aux**

Question: Use the top command to monitor the system's resource usage. What are the top five processes consuming the most CPU and memory?

Answer : **1 - P for most Cpu usage**
2 – M for the most memory usage

Bonus Question: Use the htop command (if available) for process monitoring. How does it differ from top?



Answer : htop differ from top ,, Visuals of htop have colours , options to choose from , like manual page , it shows everything with color and notions

Section 2: Control and Terminate Processes Using Signals

Question: Find the process ID (PID) of the sleep command after running sleep 600 &. How can you determine if this process is still running?

Answer : when u write **sleep 600 &**
it's pid print on the terminal , write jobs for listing of the background process and states and you can write ps aux to show the running process , ctrl+z will pausa them

Question: Use the kill command to send a SIGTERM signal to the sleep process. How can you confirm that the process has terminated?

Answer :

```
root@abdo:~# sleep 800 &
[1] 4761
root@abdo:~# jobs
[1]+  Running                 sleep 800 &
root@abdo:~# ps aux | grep 800
root         721  0.0  0.0   3740  2800 ?        Ss   12:20   0:00 /sbin/mdadm --monitor --scan
root        4761  0.0  0.0   8288  2048 pts/1    S    14:31   0:00 sleep 800
root        4763  0.0  0.0   9144  2268 pts/1    S+   14:31   0:00 grep --color=auto 800
root@abdo:~# kill -l
 1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL       5) SIGTRAP
 6) SIGABRT      7) SIGBUS       8) SIGFPE       9) SIGKILL     10) SIGUSR1
11) SIGSEGV     12) SIGUSR2     13) SIGPIPE     14) SIGALRM     15) SIGTERM
16) SIGSTKFLT   17) SIGCHLD     18) SIGCONT     19) SIGSTOP     20) SIGTSTP
21) SIGTTIN     22) SIGTTOU     23) SIGURG      24) SIGXCPU     25) SIGXFSZ
26) SIGVTALRM   27) SIGPROF     28) SIGWINCH    29) SIGIO       30) SIGPWR
31) SIGSYS      34) SIGRTMIN    35) SIGRTMIN+1  36) SIGRTMIN+2  37) SIGRTMIN+3
38) SIGRTMIN+4  39) SIGRTMIN+5  40) SIGRTMIN+6  41) SIGRTMIN+7  42) SIGRTMIN+8
43) SIGRTMIN+9  44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12 47) SIGRTMIN+13
48) SIGRTMIN+14 49) SIGRTMIN+15 50) SIGRTMAX-14 51) SIGRTMAX-13 52) SIGRTMAX-12
53) SIGRTMAX-11 54) SIGRTMAX-10 55) SIGRTMAX-9  56) SIGRTMAX-8  57) SIGRTMAX-7
58) SIGRTMAX-6  59) SIGRTMAX-5  60) SIGRTMAX-4  61) SIGRTMAX-3  62) SIGRTMAX-2
63) SIGRTMAX-1  64) SIGRTMAX
root@abdo:~# kill -15 4761
root@abdo:~# jobs | grep 800
[1]+  Terminated              sleep 800
[1]+  Terminated              sleep 800
root@abdo:~# 
```

Bonus Question: Send a SIGKILL signal to a process that is not responding. What is the effect of this signal?

Answer :

SiGKILL is the most powerfull one for closing process not responding it not request it just kill it uses : kill -9 <pid>

Section 3: Change the Priority of Running Processes

Question: Start a stress process using stress --cpu 1 --timeout 300 & in the background. What is the default priority (nice value) for this process?

Answer : the nice value it the priority of the process , the default priority is 0 is the medium priority ,
-20 is the highest priority , 19 is the lowest priority

Question: Use the renice command to lower the priority of the stress process. What effect does lowering the priority have on the system's performance?

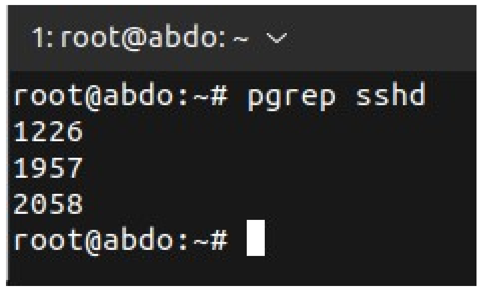
Answer : renice -n 19 <pid>
Lowering the priority of the stress process reduces its CPU scheduling preference, allowing critical and interactive processes to perform better and improving overall system responsiveness.

Bonus Question: Change the nice value of a new process when starting it with the nice command (e.g., nice -n 10 sleep 600 &). How does this affect the process?

Answer : reducing the priority in the start , will GIVE the cpu schedular less time to execute the process , shall it make the system response better

Section 4: Monitor Specific Processes

Question: Use the pgrep command to find the process ID of a running service (e.g., sshd). What information does pgrep provide?



Question: Use the pkill command to send a signal to terminate all processes with a particular name (e.g., sleep). What is the result?

Answer : **All the process named Sleep will be killed , terminated** , **not recommended to use this command as it will kill all the process naamed as u typed it may stop critical process or make undetected behaviour**

Bonus Question: Explain the difference between pkill and killall. Which scenarios would require using pkill over killall?

Answer : **`pkill`** terminates processes using pattern matching, while **`killall`** terminates processes by exact process name.

Its quitely safe to use **killall** more than **pkill**